INCH-POUND
MIL-DTL-24643/75
13 December 2011

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -55 °C TO +105 °C, 300 VOLTS, TYPE LSLCFS-25

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable is a low-capacitance foil-shielded 12½ twisted pair intended for use with computer RS232/422/485 serial transmissions and is intended to connect computers and peripheral devices from a 25-pin grounded serial interface.

Construction, Non-Watertight

First	-	Copper conductor (coated or uncoated), 24 AWG, ASTM B8, Class B stranding, 7 strands.
Second	-	Thermoset insulation. Wall thickness as necessary for the completed cable to meet the specified electrical requirements. Solid color code to be applied using Method 3; see figure 1 for colors required. Maximum diameter shall not exceed 0.055 inch.
Third	-	Two insulated conductors, twisted together such that they meet the electrical requirements specified herein. Twist lengths shall not exceed 1.5 inches.
Fourth	-	12 pairs and one insulated conductor cabled together with a lay as required to meet electrical requirements specified herein (see figure 1). Cabling sequence shall be consecutive, starting with number 1 from the center outward. Fillers are not permitted in this cable.
Fifth	-	Binder tape applied with overlap.
Sixth	-	Drain wire, copper conductor (coated or uncoated), stranded, 24 AWG, ASTM B8, Class B stranding, 7 strands.
Seventh	-	Copper foil/polyester tape. Metallic side facing in and in contact with the drain wire.
Eighth	-	Cross-linked polyolefin jacket. Minimum wall thickness of 0.040 inch.

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Color Code	
Pr 1	BLU/WHT
Pr 2	ORN/WHT
Pr 3	GRN/WHT
Pr 4	BRN/WHT
Pr 5	GRA/WHT
Pr 6	RED/BLU
Pr 7	RED/ORN
Pr 8	RED/GRN
Pr 9	RED/BRN
Pr 10	RED/GRA
Pr 11	BLK/BLU
Pr 12	BLK/ORN
Single	WHT

FIGURE 1. Illustration of construction.

TABLE I. Details.

Military part no. M24643/75	Type	Number of pairs	Overall diameter (inches)	
			min.	max.
-01UO	LSLCFS-25	12.5	0.450	0.530

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

DC resistance (ohms/1000 feet at 25 °C, max.)	25.67
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to overall shield	1000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to overall shield	500
Jacket flaws	No failure
Conductor and shield continuity	No failure

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Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance at 1 kilohertz (kHz) (pF/m, max.)	45
Between conductor and shield at 1 kHz (pF/m, max.)	80
Character impedance, at 1 megahertz (MHz) (ohms)	100±15
Group B:	
Drip (95±1 °C)	Not applicable
Cold bending, cable (when tested at -55 °C)	No failure
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Group C:	
Physicals (aged) air oven	
Insulation (when tested at 136 °C)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Shield conformance to material and application	No failure
Permanence of printing (jacket) (cycles, min.)	125

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Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
EMP response (dB, min.)	Not applicable

Group D:

Flame propagation (cable)	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Insulation	45
Toxicity index (max.)	
Jacket	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSLCFS-25	1000

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2011-033)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

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